15: Model Selection

Stat 230 - Spring 2026

> **Note**
>
> This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use https://aluby.github.io/stat230/activities/15-model-selection

> **Note**
>
> This activity goes with the Day 15 notes. Record your results in the table in your notes.

> **Tip**
>
> Run the code chunks in order: objects you create in one chunk (like a fitted model) are available in the chunks that follow.

0.1 Overview

Today you will build a model to answer this question: **How are weekends and holidays associated with ridership, after accounting for the weather?** Run the code below to load the `bikes` data and fit five candidate models.

```
bikes <- read.csv("https://aluby.github.io/stat230/data/bikes.csv")

bikes_A <- lm(rides ~ temp_actual + windspeed + weekend, data = bikes)
bikes_B <- lm(rides ~ temp_actual + windspeed + weekend + holiday, data = bikes)
bikes_C <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity, data =
bikes_D <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity + temp_f
bikes_E <- lm(rides ~ temp_feel + humidity + weekend + holiday, data = bikes)
```

0.2 Part 1: Look before you fit

Q1. Make a correlation matrix of the weather variables. Which pair of predictors could cause problems if we include both in the same model? What problem?

```
cor(bikes[, c("temp_actual", "temp_feel", "windspeed", "humidity")])
```

0.3 Part 2: Nested models and the F test

Q2. Which of these pairs of models are nested? For the pairs that are nested, which is the full model?

- A and B
- B and C
- C and D
- C and E
- A and C

The `anova()` function takes the **reduced** model first and the **full** model second.

Q3. Fill in the blanks to run the extra sums of squares F test comparing models A and B, and then models B and C. What do you conclude each time?

```
anova(___, ___)

anova(___, ___)
```

Q4. Now compare C to D (adding `temp_feel`). What does the F test suggest?

```
# your code here
```

Q5. Compare the `temp_actual` row of `summary(bikes_C)` to the `temp_actual` row of `summary(bikes_D)`. What changed, and why? How does this connect to your answer to Q1?

```
summary(bikes_C)
summary(bikes_D)
```

0.4 Part 3: Comparing models that aren't nested

Q6. Use `summary(...)$adj.r.squared`, `AIC()`, and `BIC()` to fill in the table in your notes for all five models.

```
summary(bikes_A)$adj.r.squared
AIC(bikes_A)
BIC(bikes_A)

# repeat for the other models, or use sapply:
models <- list(A = bikes_A, B = bikes_B, C = bikes_C, D = bikes_D, E = bikes_E)
sapply(models, function(m) summary(m)$adj.r.squared)
sapply(models, AIC)
sapply(models, BIC)
```

Q7. Models C and E are not nested. Which model do adjusted R^2, AIC, and BIC prefer? Why can't we use an F test to compare them?

Q8. Do the criteria agree on adding `temp_feel` to model C (comparing C and D)? What does each one say? Which model would you choose?

0.5 Part 4: Interpreting your model

Q9. Which of the five models would you use to answer our question? Justify your choice using what you found above.

Q10. Use `confint()` on your chosen model. Interpret the coefficients for `weekend` and `holiday` (and their confidence intervals) in context. Why do you think the interval for `holiday` is so wide? (*Hint:* `table(bikes$holiday)`)

```
confint(___)
```

```
table(bikes$holiday)
```

Q11. (*Extension*) Let's see what happens when we add some other candidate variables to model C. Compare C to each of the models below. Are these nested comparisons? What do the F test, adjusted R^2, AIC, and BIC say? Does adding them change your answer to our question?

```
bikes_F <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity + seasor
              data = bikes)
bikes_Y <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity + year,
              data = bikes)

# your code here
```

(*Note:* season has four levels, so adding it adds three coefficients, i.e., $d = 3$.)

0.6 Function quick reference

Function	Purpose
`lm()`	Fit a linear model
`cor()`	Correlation matrix for a set of quantitative variables
`anova(reduced, full)`	Extra sums of squares F test comparing two nested models
`summary(model)$adj.r.squared`	Adjusted R^2 for a fitted model
`AIC()`, `BIC()`	Information criteria for a fitted model (smaller is better)
`confint()`	Confidence intervals for the coefficients of a fitted model